

README: Sustainability Capstone – Decarbonizing Travel

This document serves as a guide for navigating the Sustainability Capstone project. It outlines the step-by-step workflow required to complete the analysis and provides a manifest of all included files.

Project Workflow: Step-by-Step

To complete this capstone, follow these phases:

Phase 1: Environment Setup & Data Prep

1. Create an academic team at **signup.celonis.com**
2. Register for free with the academic email address (please avoid gmail or personal accounts)
3. Open the invitation e-mail and complete the registration as **STUDENT**
4. Access your academic account

Phase 2: Data Modeling

1. Data Pool Integration: Upload the CSV files into a Celonis Data Pool.
2. Create Relationships: Link the tables to allow for granular filtering. The **TripID** is the common key across the travel data, event log, and attributes.
 - *Minimum Requirement:* Ensure the **CarbonEmission_kgCO2e** field is correctly mapped.

Phase 3: Process Analysis & Dashboarding

- ❖ Process Discovery: Build a Process Explorer and Variant Explorer using the **public_trip_event_log.csv**. Identify bottlenecks or "out-of-policy" behaviors.
- ❖ Build Views: Create at least three distinct dashboard tabs focusing on:
 - Emissions Overview: High-level KPIs (Total CO2, Costs).
 - Benchmarking: Emissions by Region or Business Unit.
 - Emission Drilldown: With actionable Insights: Identifying specific metrics (e.g., Flight vs. Train).

Phase 4: Final Pitch

1. Synthesize Findings: Identify "Hotspots" (high-emission areas) and "Quick Wins."
2. Develop Recommendations: Propose specific policy updates (e.g., "All trips under X km must be by train").
3. Present: Deliver a consultant-style pitch including a live demo of your Celonis environment.

File Manifest & Descriptions

Core Datasets (CSV)

- [public_trip_data.csv](#): The master data table. Contains trip details (locations, transport type, costs) and pre-calculated carbon emissions. Use this for the standard project path.
- [public_trip_event_log.csv](#): The activity table. Contains timestamps for every step of a trip (Booking -> Approval -> Departure -> Reimbursement). Essential for the Process Explorer.
- [public_trip_event_attributes.csv](#): Contextual data. Provides specific reasons for policy exceptions, flight cancellations, or hotel changes. Use this to enrich your analysis of why certain process variants occur.

Documentation

- [Sustainability Capstone Outline: Decarbonizing Travel](#) Provides the business case, the "Net Zero by 2030" goal, and the evaluation criteria for your final presentation.
- [<FullName>_Project](#) Use this template to submit your presentation.

Learning Materials

Recommendations from the Celonis Academy:

[Introduction to Process Mining](#)

[Process Mining Fundamentals](#)

[Rising Stars Business and Technical Tracks](#)

[Sustainability Bootcamp](#) - to learn about sustainability and our current solutions

Collaboration Examples:

[Climatiq x Celonis collaborations](#)

Celonis AI Foundations

[Configure Process Copilots](#)

[Accelerate Process Analysis with Insight Explorer](#)

[Maximize Data Value with AI Annotation Builder](#)